SRM VEC

SRM Nagar, Kattankulathur – 603 203

DEPARTMENT OF AIDS

QUESTION BANK

SUBJECT: AD3445 – AI1

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UNIT I – INTRODUCTION AND PROBLEM SOLVING

Introduction to AI, Intelligent Agents, Problem Solving Agents, State Space Search, Uninformed Search Strategies: Breadth-first Search, Depth-first Search, Uniform Cost Search, Depth Limited Search, Iterative Deepening Search, Bidirectional Search

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define Artificial Intelligence based on the 'Acting Rationally' approach.	BTL1	Remembering
2	What is the Turing Test, and what is its primary purpose in AI?	BTL1	Remembering
3	List the four components of a PEAS description for an intelligent agent.	BTL1	Remembering
4	Distinguish between a fully observable and a partially observable environment.	BTL2	Understanding
5	Define a rational agent and explain how rationality is measured.	BTL1	Remembering
6	What are the five basic types of agent programs?	BTL1	Remembering
7	Identify the four criteria used to evaluate the performance of a search algorithm.	BTL1	Remembering
8	Explain the concept of 'State Space' in the context of problem-solving agents.	BTL2	Understanding
9	Why is Breadth-First Search (BFS) considered optimal for problems with constant step costs?	BTL2	Understanding
10	State the time and space complexity of Depth-First Search (DFS).	BTL1	Remembering
11	How does Uniform Cost Search (UCS) differ from Breadth-First Search?	BTL2	Understanding
12	What is the primary motivation behind using Depth Limited Search (DLS)?	BTL2	Understanding
13	Define the term 'branching factor' in a search tree.	BTL1	Remembering
14	Why is Iterative Deepening Search (IDS) preferred for large state spaces with unknown goal depth?	BTL2	Understanding
15	Explain the basic idea of Bidirectional Search.	BTL2	Understanding
16	What is the difference between a 'state' and a 'node' in search terminology?	BTL2	Understanding
17	Define 'Problem Formulation' in the design of a problem-solving agent.	BTL1	Remembering
18	What is an 'episodic' environment in AI?	BTL1	Remembering
19	Mention one significant disadvantage of Bidirectional Search.	BTL2	Understanding
20	What is a 'Simple Reflex Agent'?	BTL1	Remembering

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the four different approaches to Artificial Intelligence: Thinking Humanly, Thinking Rationally, Acting Humanly, and Acting Rationally.	(16)	BTL4	Analyzing
2	Design a complete PEAS description and characterize the environment for an Autonomous Mars Rover and a Crossword Puzzle Solver.	(16)	BTL6	Creating
3	Compare and contrast the architectures of Model-based Reflex Agents and Utility-based Agents with neat diagrams.	(16)	BTL4	Analyzing
4	Formulate the 8-puzzle problem as a state-space search problem, specifying the states, initial state, actions, transition model, and goal test.	(16)	BTL3	Applying
5	Describe the general 'Search-and-Execute' framework used by problem-solving agents and explain the importance of abstraction in problem formulation.	(16)	BTL3	Applying
6	Provide a detailed comparative analysis of BFS, DFS, and UCS based on completeness, optimality, time complexity, and space complexity.	(16)	BTL4	Analyzing

7	Illustrate the step-by-step execution of Uniform Cost Search on a graph with varying edge costs and prove its optimality conditions.	(16)	BTL3	Applying
8	Critically evaluate the performance of Iterative Deepening Search (IDS). Why is it often considered the preferred uninformed search method for large search spaces?	(16)	BTL5	Evaluating
9	Explain the mechanism of Bidirectional Search. Derive its time complexity and discuss the practical difficulties in its implementation.	(16)	BTL4	Analyzing
10	Analyze the impact of different environment properties (Deterministic vs. Stochastic, Static vs. Dynamic, Discrete vs. Continuous) on agent design.	(16)	BTL4	Analyzing
11	Design a Goal-based Agent for an automated warehouse robot. Explain how it differs from a simple reflex agent in terms of decision making.	(16)	BTL6	Creating
12	Demonstrate the execution of Depth Limited Search on a tree of depth 10 with a limit L=4. Discuss what happens if the goal is at depth 6.	(16)	BTL3	Applying
13	Evaluate the Turing Test as a measure of machine intelligence. Discuss its strengths and the common criticisms raised against it.	(16)	BTL5	Evaluating
14	Derive the total number of nodes generated by Iterative Deepening Search in the worst case and compare it with the number of nodes in BFS.	(16)	BTL4	Analyzing
15	Explain the 'Vacuum-World' problem. Formulate it as a state space search and show the state space graph for a two-room configuration.	(16)	BTL3	Applying
16	Justify the use of Utility-based agents in environments where goals are not enough to make a high-quality decision, such as automated stock trading.	(16)	BTL5	Evaluating
17	Construct a search tree for the 'Missionaries and Cannibals' problem and explain how a search algorithm would find a solution.	(16)	BTL6	Creating
18	Analyze the memory limitations of Breadth-First Search and explain how Depth-First Search manages to be more memory-efficient.	(16)	BTL4	Analyzing
19	Compare the 'Atomic', 'Factored', and 'Structured' representations of states used by intelligent agents.	(16)	BTL4	Analyzing
20	Evaluate the trade-offs between time complexity and space complexity in search algorithms. Which is usually the more critical constraint in AI?	(16)	BTL5	Evaluating

UNIT II – INFORMED SEARCH AND GAME PLAYING

Heuristic Search, A Search, AO* Search, Local Search Algorithms, Hill Climbing, Simulated Annealing, Genetic Algorithms, Adversarial Search, Game Playing, Minimax Algorithm, Alpha-Beta Pruning*

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define a heuristic function and its role in informed search strategies.	BTL1	Remembering
2	State the condition for a heuristic to be considered admissible in A* search.	BTL1	Remembering
3	Distinguish between informed search and uninformed search with an example.	BTL2	Understanding
4	What is the significance of the 'f(n) = g(n) + h(n)' formula in A* search?	BTL2	Understanding
5	Define an AND-OR graph and identify where it is primarily used.	BTL1	Remembering
6	Explain the concept of 'Local Optima' in the context of Hill Climbing.	BTL2	Understanding
7	What is the role of the 'temperature' parameter in the Simulated Annealing algorithm?	BTL2	Understanding
8	List the three main genetic operators used in Genetic Algorithms.	BTL1	Remembering
9	Define 'Fitness Function' and its importance in evolutionary search.	BTL1	Remembering
10	What is a 'Zero-Sum' game in adversarial search?	BTL1	Remembering
11	Explain the purpose of the Alpha and Beta values in Alpha-Beta pruning.	BTL2	Understanding
12	Under what condition does Alpha-Beta pruning provide the maximum search space reduction?	BTL2	Understanding
13	Define the 'Horizon Effect' in game playing algorithms.	BTL1	Remembering
14	What is a 'Consistent' (or Monotonic) heuristic?	BTL1	Remembering
15	Briefly explain the 'Plateau' problem in Hill Climbing search.	BTL2	Understanding
16	How does AO* search differ from A* search in terms of the goal state?	BTL2	Understanding
17	Define 'Ply' and 'Branching Factor' in the context of game trees.	BTL1	Remembering
18	What is the 'Crossover' operation in Genetic Algorithms?	BTL2	Understanding
19	State the Minimax decision rule for a MAX player.	BTL1	Remembering
20	Identify the main drawback of the pure Hill Climbing algorithm.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the A* search algorithm in detail. Prove that A* is optimal if the heuristic function is admissible.	(16)	BTL4	Analyzing
2	Describe the AO* algorithm for searching AND-OR graphs. Illustrate its working with a suitable example of problem decomposition.	(16)	BTL3	Applying
3	Compare and contrast Hill Climbing, Steepest-Ascent Hill Climbing, and Stochastic Hill Climbing. Discuss how they handle local maxima.	(16)	BTL4	Analyzing
4	Explain the Simulated Annealing process. How does the probability of accepting a worse move change over time, and why is this beneficial?	(16)	BTL4	Analyzing
5	Design a Genetic Algorithm to solve the 8-Queens problem. Specify the chromosome representation, fitness function, and selection mechanism.	(16)	BTL6	Creating
6	Trace the Minimax algorithm for a game tree of depth 3 with a branching factor of 2. Explain how the values are backed up to the root.	(16)	BTL3	Applying
7	Demonstrate the Alpha-Beta pruning process on a given game tree. Show the nodes pruned and explain the conditions that triggered the pruning.	(16)	BTL3	Applying

8	Evaluate the impact of node ordering on the efficiency of Alpha-Beta pruning. What is the time complexity in the best and worst cases?	(16)	BTL5	Evaluating
9	Analyze the trade-offs between using a simple heuristic versus a complex, computationally expensive heuristic in A* search.	(16)	BTL4	Analyzing
10	Discuss the 'Exploration vs. Exploitation' trade-off in Genetic Algorithms. How do mutation and crossover rates influence this balance?	(16)	BTL4	Analyzing
11	Develop a heuristic function for the 15-puzzle problem. Justify whether your heuristic is admissible and consistent.	(16)	BTL6	Creating
12	Explain the concept of 'Heuristic Path Algorithm' and how it generalizes both Breadth-First Search and Dijkstra's algorithm.	(16)	BTL4	Analyzing
13	Critically examine the limitations of the Minimax algorithm in games with high branching factors like Go or Chess. Suggest potential improvements.	(16)	BTL5	Evaluating
14	Propose a modified version of Hill Climbing that incorporates a 'tabu list' to avoid cycles. Explain how this improves search performance.	(16)	BTL6	Creating
15	Compare the memory requirements and completeness of A* search with Iterative Deepening A* (IDA*).	(16)	BTL4	Analyzing
16	Explain how Simulated Annealing can be viewed as a search that mimics the physical process of cooling metals. Derive the acceptance probability formula.	(16)	BTL4	Analyzing
17	Design an evaluation function for a game of Tic-Tac-Toe. How would this function be used by a Minimax agent to choose the next move?	(16)	BTL6	Creating
18	Evaluate the performance of Genetic Algorithms in solving the Traveling Salesperson Problem (TSP) compared to traditional local search methods.	(16)	BTL5	Evaluating
19	Analyze the behavior of the A* algorithm when the heuristic function h(n) is always zero. Which uninformed search does it become?	(16)	BTL4	Analyzing
20	Formulate a search strategy for a multi-agent adversarial environment where players do not have perfect information. Contrast this with standard Minimax.	(16)	BTL6	Creating

UNIT III – KNOWLEDGE REPRESENTATION AND LOGIC

Knowledge-Based Agents, Propositional Logic, First-Order Logic, Inference in FOL, Forward Chaining, Backward Chaining, Resolution, Unification, Knowledge Representation Issues

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define a Knowledge-Based Agent and its two primary components.	BTL1	Remembering
2	Distinguish between the 'declarative' and 'procedural' approaches to building an agent.	BTL2	Understanding
3	What is meant by 'entailment' in the context of formal logic?	BTL1	Remembering
4	Define 'soundness' and 'completeness' for an inference algorithm.	BTL1	Remembering
5	Explain the difference between a 'valid' sentence and a 'satisfiable' sentence.	BTL2	Understanding
6	State the purpose of the 'Wumpus World' environment in AI studies.	BTL1	Remembering
7	List the basic elements of First-Order Logic (FOL) syntax.	BTL1	Remembering
8	What is a 'Most General Unifier' (MGU) in the context of unification?	BTL2	Understanding
9	Define 'Skolemization' and its role in logic conversion.	BTL1	Remembering
10	What is a 'Horn Clause' and why is it significant for inference?	BTL2	Understanding
11	Briefly explain the 'Frame Problem' in knowledge representation.	BTL2	Understanding
12	Define the 'Resolution' rule of inference.	BTL1	Remembering
13	What is the difference between a 'constant' and a 'function' symbol in FOL?	BTL2	Understanding
14	Define 'Forward Chaining' in the context of production systems.	BTL1	Remembering
15	What is 'Backward Chaining' and how does it differ from Forward Chaining?	BTL2	Understanding
16	List two major issues in Knowledge Representation (KR).	BTL1	Remembering
17	Explain the concept of 'Model' in Propositional Logic.	BTL2	Understanding
18	What is 'Inferential Adequacy' in a knowledge representation scheme?	BTL1	Remembering
19	Define 'Universal Quantification' and 'Existential Quantification'.	BTL1	Remembering
20	Explain the 'Closed World Assumption' (CWA) in logic-based systems.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Describe the general architecture of a Knowledge-Based Agent. Illustrate how such an agent functions within the Wumpus World environment, detailing the TELL and ASK operations.	(16)	BTL3	Applying
2	Compare Propositional Logic and First-Order Logic. Analyze why Propositional Logic is insufficient for representing complex domains and how FOL addresses these limitations.	(16)	BTL4	Analyzing
3	Explain the process of converting a First-Order Logic sentence into Conjunctive Normal Form (CNF). Provide a step-by-step derivation for a complex sentence involving quantifiers and implications.	(16)	BTL3	Applying
4	Detailed the Unification algorithm. Given the literals P(a, x, f(g(y))) and P(z, h(z, w), f(w)), demonstrate the step-by-step execution of the algorithm to find the MGU.	(16)	BTL3	Applying
5	Critically evaluate the Resolution Refutation method in FOL. Prove the statement 'Socrates is mortal' given 'All men are mortal' and 'Socrates is a man' using resolution.	(16)	BTL5	Evaluating
6	Analyze the Forward Chaining algorithm for definite clauses. Discuss its data-driven nature and illustrate its execution with a sample knowledge base and a target goal.	(16)	BTL4	Analyzing

7	Examine the Backward Chaining algorithm. Explain its goal-directed nature and discuss the potential issues of redundant computations and infinite loops in logic programming.	(16)	BTL4	Analyzing
8	Design a First-Order Logic Knowledge Base for a 'Smart Library System' that handles book loans, member categories, and overdue fines. Include at least five distinct rules.	(16)	BTL6	Creating
9	Discuss the various issues in Knowledge Representation, specifically focusing on Granularity, Set of Objects, and the Qualification problem. Provide examples for each.	(16)	BTL4	Analyzing
10	Explain the Generalized Modus Ponens (GMP). Justify why GMP is more efficient than standard Modus Ponens when dealing with First-Order Logic inference.	(16)	BTL5	Evaluating
11	Analyze the semantics of nested quantifiers in FOL. Evaluate how the swapping of universal and existential quantifiers changes the meaning of a sentence with concrete examples.	(16)	BTL4	Analyzing
12	Develop a resolution-based proof to solve the 'Crime Investigation' problem: If the suspect is guilty, he was at the scene. If he was at the scene, he has a motive. Prove guilt given motive and scene presence.	(16)	BTL6	Creating
13	Evaluate the trade-offs between 'Model Checking' and 'Theorem Proving' as strategies for logical inference in large-scale knowledge bases.	(16)	BTL5	Evaluating
14	Explain the concept of 'Representational Adequacy' and 'Inferential Efficiency'. Discuss how these two factors conflict when choosing a logic-based representation.	(16)	BTL4	Analyzing
15	Implement a logical representation for the 'Kinship' domain. Define predicates for parent, sibling, grandparent, and cousin using FOL and show how 'grandparent' is inferred.	(16)	BTL6	Creating
16	Compare the efficiency of Forward Chaining and Backward Chaining in terms of time complexity and space complexity for a tree-structured knowledge base.	(16)	BTL4	Analyzing
17	Justify the use of 'Equality' in FOL. Explain the 'Paramodulation' and 'Demodulation' techniques used to handle equality in automated reasoning systems.	(16)	BTL5	Evaluating
18	Construct a Knowledge-Based Agent for the 'Electronic Circuits' domain. Describe how the agent can diagnose a faulty gate using logical inference.	(16)	BTL6	Creating
19	Analyze the 'Ramification Problem' and 'Modification Problem' in the context of dynamic environments where facts change over time.	(16)	BTL4	Analyzing
20	Critique the use of First-Order Logic for representing 'Common Sense' knowledge. Suggest how extensions to FOL might handle exceptions and defaults.	(16)	BTL5	Evaluating

UNIT IV – KNOWLEDGE REPRESENTATION AND PLANNING

Ontological Engineering, Categories and Objects, Events, Mental Objects and Belief Systems, Reasoning with Default Information, Classical Planning, Planning as State-Space Search, Planning Graphs

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define Ontological Engineering and its significance in AI.	BTL1	Remembering
2	Distinguish between a general-purpose ontology and a domain-specific ontology.	BTL2	Understanding
3	What is meant by 'Subsumption' in the context of categories?	BTL1	Remembering
4	Explain the concept of 'Fluents' in situation calculus.	BTL2	Understanding
5	What is the 'Frame Problem' in knowledge representation?	BTL1	Remembering
6	Define 'Reification' and provide a simple example of its use.	BTL2	Understanding
7	What are 'Mental Objects' in the context of belief systems?	BTL1	Remembering
8	Explain the 'Referential Transparency' issue in belief logic.	BTL2	Understanding
9	Define Non-monotonic reasoning.	BTL1	Remembering
10	What is 'Circumscription' in default reasoning?	BTL2	Understanding
11	List the four components of a classical planning problem.	BTL1	Remembering
12	Differentiate between the Open World Assumption and the Closed World Assumption.	BTL2	Understanding
13	What is 'Progression Planning' in state-space search?	BTL1	Remembering
14	Define 'Regression Planning' and state its primary advantage.	BTL2	Understanding
15	What is a 'Planning Graph'?	BTL1	Remembering
16	Explain the concept of 'Mutex' (Mutual Exclusion) in planning graphs.	BTL2	Understanding
17	What does it mean for a planning graph to 'Level Off'?	BTL1	Remembering
18	Define the 'Qualification Problem' in action representation.	BTL1	Remembering
19	State the difference between 'Actions' and 'Events' in ontological terms.	BTL2	Understanding
20	What is a 'Literal' in PDDL (Planning Domain Definition Language)?	BTL1	Remembering

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Elaborate on the process of Ontological Engineering. Discuss the challenges involved in creating a comprehensive Upper Ontology for diverse domains.	(16)	BTL4	Analyzing
2	Demonstrate how First-Order Logic can be used to represent Categories, Measures, and Composite Objects. Provide a formal representation for a 'Smart Warehouse' inventory.	(16)	BTL3	Applying
3	Analyze the Situation Calculus framework. Explain how it handles the representation of change, and discuss its solutions to the Frame, Ramification, and Qualification problems.	(16)	BTL4	Analyzing
4	Evaluate the use of Modal Logic for representing Mental Objects and Belief Systems. How does it resolve the issue of logical omniscience?	(16)	BTL5	Evaluating
5	Compare and contrast Default Logic and Circumscription as formalisms for non-monotonic reasoning. Provide examples where one is preferred over the other.	(16)	BTL4	Analyzing
6	Design a complete PDDL domain and problem description for a 'Robot Courier' that must move packages between rooms, considering battery constraints.	(16)	BTL6	Creating
7	Analyze the efficiency of Forward State-Space Search versus Backward State-Space Search. Under what conditions does Regression Planning outperform Progression?	(16)	BTL4	Analyzing

8	Construct a Planning Graph for a simple 'Spare Tire' problem. Detail the literal levels, action levels, and the derivation of mutex relations.	(16)	BTL6	Creating
9	Explain the GraphPlan algorithm in detail. Discuss how it uses the planning graph to extract a valid plan and how it handles backtracking.	(16)	BTL3	Applying
10	Evaluate the various heuristics derived from Planning Graphs, such as Max-level, Level-sum, and Set-level heuristics, for their admissibility and effectiveness.	(16)	BTL5	Evaluating
11	Develop a knowledge representation schema for 'Time and Intervals' using Allen’s Interval Algebra. Show how it can be used to reason about overlapping events.	(16)	BTL6	Creating
12	Discuss the 'Blocks World' problem in classical planning. Formulate the STRIPS operators and show the step-by-step state transitions for a specific goal configuration.	(16)	BTL3	Applying
13	Analyze the impact of the 'Closed World Assumption' on the complexity of reasoning in large-scale knowledge bases. Contrast this with the 'Unique Names Assumption'.	(16)	BTL4	Analyzing
14	Critique the use of Semantic Networks versus Description Logics for representing taxonomic knowledge. Which is more suitable for automated reasoning and why?	(16)	BTL5	Evaluating
15	Design a reasoning system that uses 'Default Information' to handle exceptions in a biological taxonomy (e.g., 'Birds fly, but ostriches do not').	(16)	BTL6	Creating
16	Explain the concept of 'Planning as Satisfiability' (SatPlan). How are planning problems translated into propositional logic formulas?	(16)	BTL3	Applying
17	Analyze the role of 'Heuristic Search' in planning. Explain the 'Ignore Preconditions' and 'Independent Subgoals' heuristics with suitable examples.	(16)	BTL4	Analyzing
18	Evaluate the limitations of Classical Planning when applied to real-world environments characterized by uncertainty and exogenous events.	(16)	BTL5	Evaluating
19	Construct a formal representation for 'Knowledge and Belief' using Kripke structures (Possible Worlds Semantics). Explain how accessibility relations define belief.	(16)	BTL6	Creating
20	Compare the expressiveness of PDDL with earlier languages like STRIPS and ADL. Discuss how PDDL handles conditional effects and universal quantification.	(16)	BTL4	Analyzing

UNIT V – UNCERTAIN KNOWLEDGE AND REASONING

Acting under Uncertainty, Basic Probability Notation, Bayes' Rule, Probabilistic Reasoning, Bayesian Networks, Semantics of Bayesian Networks, Exact Inference in Bayesian Networks, Approximate Inference, Causal Networks

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define uncertainty in the context of Artificial Intelligence.	BTL1	Remembering
2	State the product rule and the sum rule of probability.	BTL1	Remembering
3	What is the difference between prior probability and posterior probability?	BTL2	Understanding
4	Write the mathematical expression for Bayes' Rule and identify its components.	BTL1	Remembering
5	Define a Bayesian Network and list its two main components.	BTL1	Remembering
6	Explain the concept of conditional independence between two random variables.	BTL2	Understanding
7	What is a Joint Probability Distribution (JPD)?	BTL1	Remembering
8	Define the Markov condition in a Bayesian Network.	BTL2	Understanding
9	What is meant by 'Exact Inference' in probabilistic reasoning?	BTL1	Remembering
10	List two common algorithms used for approximate inference in Bayesian Networks.	BTL1	Remembering
11	Explain the role of a Conditional Probability Table (CPT) in a Bayesian Network.	BTL2	Understanding
12	Define a Causal Network and how it differs from a simple correlation.	BTL1	Remembering
13	What is the purpose of normalization in the context of Bayes' Rule?	BTL2	Understanding
14	Define 'Evidence' and 'Query' variables in the context of inference.	BTL1	Remembering
15	What is Rejection Sampling in approximate inference?	BTL1	Remembering
16	Explain the concept of Likelihood Weighting.	BTL2	Understanding
17	Why is exact inference computationally difficult (NP-hard) in general networks?	BTL2	Understanding
18	Define the term 'd-separation' in Bayesian Networks.	BTL1	Remembering
19	What is a Directed Acyclic Graph (DAG) in the context of Bayesian Networks?	BTL1	Remembering
20	Explain the significance of the 'Chain Rule' for Bayesian Networks.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the process of acting under uncertainty and justify why classical logic fails in uncertain domains.	(16)	BTL5	Evaluating
2	Derive the Bayes' Rule from the product rule and demonstrate its application in a medical diagnosis scenario where a test result is given.	(16)	BTL3	Applying
3	Describe the semantics of Bayesian Networks and explain how they represent a full joint probability distribution using the local semantics approach.	(16)	BTL4	Analyzing
4	Design a Bayesian Network for a 'Smart Home Alarm System' involving variables: Burglary, Earthquake, Alarm, JohnCalls, and MaryCalls. Define the network structure and provide sample CPTs.	(16)	BTL6	Creating
5	Explain the Inference by Enumeration algorithm in detail and analyze its time and space complexity for a network with 'n' Boolean variables.	(16)	BTL4	Analyzing
6	Discuss the Variable Elimination algorithm for exact inference and illustrate its steps using a specific query on a small network.	(16)	BTL3	Applying

7	Compare and contrast Rejection Sampling and Likelihood Weighting. Evaluate which method is more efficient when evidence is rare.	(16)	BTL5	Evaluating
8	Explain the Gibbs Sampling algorithm and describe how it uses Markov Chain Monte Carlo (MCMC) for approximate inference.	(16)	BTL3	Applying
9	Analyze the role of Causal Networks in AI. How does a causal model differ from a purely diagnostic model in terms of structure and reasoning?	(16)	BTL4	Analyzing
10	Develop a complete probabilistic model for a 'Car Starting' problem involving Battery, Fuel, SparkPlugs, and Starter. Formulate the conditional dependencies.	(16)	BTL6	Creating
11	Evaluate the impact of node ordering on the compactness of a Bayesian Network. Provide an example where a poor ordering leads to a more complex graph.	(16)	BTL5	Evaluating
12	Explain the concept of 'Independence' and 'Conditional Independence' using basic probability notation and prove how they simplify the joint distribution.	(16)	BTL4	Analyzing
13	Describe the process of constructing a Bayesian Network from a set of variables and explain how to determine if the resulting structure is optimal.	(16)	BTL3	Applying
14	Analyze the 'Polytree' structure in Bayesian Networks and explain why exact inference is efficient in such networks compared to multiply connected networks.	(16)	BTL4	Analyzing
15	Justify the use of approximate inference methods over exact inference for large-scale real-world applications like weather forecasting.	(16)	BTL5	Evaluating
16	Explain the d-separation criterion in detail and use it to identify all independent variables in a given complex Bayesian Network structure.	(16)	BTL4	Analyzing
17	Formulate the mathematical steps for calculating the posterior distribution P(Query | Evidence) using the Likelihood Weighting algorithm.	(16)	BTL3	Applying
18	Create a comparative study of different sampling methods (Direct, Rejection, Likelihood, Gibbs) used in probabilistic reasoning.	(16)	BTL6	Creating
19	Explain how Bayesian Networks can be used for both diagnostic reasoning (bottom-up) and causal reasoning (top-down).	(16)	BTL3	Applying
20	Analyze the relationship between the topology of a Bayesian Network and the conditional independence assumptions it encodes.	(16)	BTL4	Analyzing

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